host_agent

Initialised

User: Show me all houses in Exeter that are not within 2km of a library.

Calling tool `send_message` with args:

```
{'target': 'planning_agent', 'task_description': 'User wants to find all houses in Exeter that are not within 2km of a library. Provide the general steps to s
```

planning_agent

Initialised

1. The general geographical area is Exeter, so there will be 1 entry for Exeter. 2. The entities to search for are houses in Exeter, so we need to find as many entries as possible for houses. 3. We also need to identify libraries in Exeter, which will be 1 entry for libraries. 4. The condition is to find houses that are not within 2km of any library in Exeter.

output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Find libraries in Exeter as many search results", "Apply condition to exclude houses within 2km of any library"]

host_agent

Tool send_message returned result:

1. The general geographical area is Exeter, so there will be 1 entry for Exeter. 2. The entities to search for are houses in Exeter, so we need to find as many entries as possible for houses. 3. We also need to identify libraries in Exeter, which will be 1 entry for libraries. 4. The condition is to find houses that are not within 2km of any library in Exeter.

output steps: ["Find Exeter 1 area", "Find houses in Exeter as many search results", "Find libraries in Exeter as many search results", "Apply condition to exclude houses within 2km of any library"]

Calling tool `send_message` with args:

```
{'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}
```

---

named_area_agent

Initialised

Calling tool `call_os_ngd` with args:

```
{'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}
```

Tool call_os_ngd returned result: - Search results have been found. - Artifact generated is : exeter_area_polygon - Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter.. - Count of records fetched : 98.

Calling tool `send_message` with args:

```
{'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filter
```

---

data_analysis_agent

Initialised

Calling tool `generate_metadata_for_artifacts` with args:

```
{'artifact_names': ['exeter_area_polygon']}
```

Tool generate_metadata_for_artifacts returned result: (metadata summary omitted)

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
```

```python
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using all relevant columns
    filtered = gdf[
        (
            (gdf['name1_text'].str.lower() == 'exeter') |
            (gdf['name1_text'].str.lower().str.contains('exeter'))
        ) & (
            (gdf['descriptiongroup'].str.lower().isin(['settlement', 'part of settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town', 'suburban area']))
        )
    ]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    return [
        summary,
        'filtered_exeter_area_polygon',
        'GeoDataFrame containing only the polygon(s) for the Exeter area, filtered by name and settlement/city/town/suburban area description.',
        filtered
    ]
```

Tool code_executor returned result: - Filtered to 1 polygon(s) for Exeter area.

Output from agent data_analysis_agent: Filtered to 1 polygon for the Exeter area. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for the Exeter area, filtered by name and relevant settlement/city/town/suburban area description.

Tool send_message returned result: - Filtered to 1 polygon for the Exeter area. The result is saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for the Exeter area, filtered by name and relevant settlement/city/town/suburban area description.

---

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This file contains only the polygon(s) for the Exeter area, filtered by name and relevant settlement/city/town/suburban area description.

Tool send_message returned result: - The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'.

---

host_agent

Calling tool `send_message` with args:

```
{'target': 'buildings', 'task_description': "Find all houses in the area defined by 'filtered_exeter_area_polygon'. Filter for residential houses only."}
```

---

buildings_agent

Initialised

Calling tool `call_os_ngd` with args:

```
{'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'all_buildings_in_filtered_exeter_area
```

Tool call_os_ngd returned result: - Multiple search results have been found in multiple datasets. A summary of each is provided. - Artifacts generated are : ['buildings_all_buildings_in_filtered_exeter_area.geojson', 'buildingline_all_buildings_in_filtered_exeter_area.geojson', 'buildingpart_all_buildings_in_filtered_exeter_area.geojson'] - Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Find all houses in the area defined by 'filtered_exeter_area_polygon'. Filter for residential houses only. using filters None.", ...] - Counts of records fetched are : [69169, 69342, 85480].

Calling tool `send_message` with args:

```
{'target': 'data_analysis_agent', 'task_description': "From 'buildings_all_buildings_in_filtered_exeter_area.geojson', filter and return only residential hous
```

---

data_analysis_agent

Calling tool `generate_metadata_for_artifacts` with args:

```
{'artifact_names': ['buildings_all_buildings_in_filtered_exeter_area.geojson']}
```

Tool generate_metadata_for_artifacts returned result: (metadata summary omitted)

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_residential_houses(data: list):
```

```
    gdf = data[0]
    # Filter for residential houses: buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total == 1
    filtered = gdf[(gdf['buildinguse_addresscount_residential'] > 0) & (gdf['buildinguse_addresscount_total'] == 1)]
    # Save to geojson
    filtered.to_file('residential_houses_in_filtered_exeter_area.geojson', driver='GeoJSON')
    summary = f"Filtered {len(filtered)} residential houses from {len(gdf)} total buildings."
    description = "GeoDataFrame containing only residential houses (where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total == 1) in
    return [summary, 'residential_houses_in_filtered_exeter_area.geojson', description, filtered]
```

Tool code_executor returned result: - Filtered 38848 residential houses from 69169 total buildings.

Output from agent data_analysis_agent: Filtered 38,848 residential houses from 69,169 total buildings. The results are saved as 'residential_houses_in_filtered_exeter_area.geojson', which contains only residential houses (where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total == 1) in the filtered Exeter area.

Tool send_message returned result: - Filtered 38,848 residential houses from 69,169 total buildings. The results are saved as 'residential_houses_in_filtered_exeter_area.geojson'.

---

buildings_agent

I have filtered all buildings in the area defined by 'filtered_exeter_area_polygon' to return only residential houses. A house is defined as a building where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1.

Results: - 38,848 residential houses were found out of 69,169 total buildings. - The filtered results are saved as: residential_houses_in_filtered_exeter_area.geojson

Artifact description: - 'residential_houses_in_filtered_exeter_area.geojson': GeoDataFrame containing only residential houses (where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total == 1) in the filtered Exeter area.

Let me know if you need further analysis or details on these houses!

Tool send_message returned result: - I have filtered all buildings in the area defined by 'filtered_exeter_area_polygon' to return only residential houses. ...

---

host_agent

Calling tool `send_message` with args:

```
{'target': 'land_use_features', 'task_description': "Find all libraries in the area defined by 'filtered_exeter_area_polygon'."}
```

---

land_use_features_agent

Initialised

Calling tool `call_os_ngd` with args:

```
{'filters': ['Library'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'libraries_in_exeter'}
```

Tool call_os_ngd returned result: - Multiple search results have been found in multiple datasets. A summary of each is provided. - Artifacts generated are : ['landuse_libraries_in_exeter'] - Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Find all libraries in the area defined by 'filtered_exeter_area_polygon'. using filters ['Library']. Further Name filtering is available for this"] - Counts of records fetched are : [3].

I found 3 libraries in the area defined by 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'landuse_libraries_in_exeter'.

Tool send_message returned result: - I found 3 libraries in the area defined by 'filtered_exeter_area_polygon'. The data is saved in the artifact named 'landuse_libraries_in_exeter'.

---

host_agent

Calling tool `generate_metadata_for_all_artifacts` with args:

```
{}
```

Tool generate_metadata_for_all_artifacts returned result: (artifact descriptions summary omitted)

Calling tool `send_message` with args:

```
{'target': 'plotting_agent', 'task_description': "Plot all residential houses from 'residential_houses_in_filtered_exeter_area.geojson' in Exeter that are NOT
```

plotting_agent

Initialised

Calling tool `generate_metadata_for_artifacts` with args:

```
{'artifact_names': ['residential_houses_in_filtered_exeter_area.geojson', 'landuse_libraries_in_exeter']}
```

Tool generate_metadata_for_artifacts returned result: (metadata summary omitted)

Tool Code Executor

```python
def plot_residential_houses_not_within_2km_of_library(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import Polygon
    from shapely.ops import unary_union

    # Unpack data
    houses = data[0].copy()
    libraries = data[1].copy()

    # Filter for residential houses only (use both buildinguse and description for quality)
    residential_houses = houses[
        (houses['buildinguse'] == 'Residential Accommodation') &
        (houses['description'].str.contains('House', case=False, na=False)) &
        (houses['physicalstate'] == 'Built')
    ].copy()

    # Ensure both are in EPSG:4326 for folium and distance calculations
    residential_houses = residential_houses.to_crs(epsg=4326)
    libraries = libraries.to_crs(epsg=4326)

    # Project to a metric CRS for buffer (use British National Grid: EPSG:27700)
    residential_houses_metric = residential_houses.to_crs(epsg=27700)
    libraries_metric = libraries.to_crs(epsg=27700)

    # Create a 2km buffer around all libraries (unary_union for all libraries)
    libraries_union = unary_union(libraries_metric['geometry'])
    buffer_2km = libraries_union.buffer(2000)

    # Find houses NOT within 2km of any library
    not_within_2km = residential_houses_metric[~residential_houses_metric.intersects(buffer_2km)].copy()
```

```python
# Convert back to EPSG:4326 for plotting
not_within_2km = not_within_2km.to_crs(epsg=4326)
buffer_2km_wgs84 = gpd.GeoSeries([buffer_2km], crs=27700).to_crs(epsg=4326)

# Prepare folium map
if not not_within_2km.empty:
    centroid = not_within_2km.geometry.iloc[0].centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12)
else:
    # Default to Exeter city center if no results
    m = folium.Map(location=[50.7218, -3.5336], zoom_start=12)

# Plot all libraries
for _, row in libraries.iterrows():
    folium.GeoJson(row['geometry'], name='Library', style_function=lambda x: {'color': 'blue', 'fillOpacity': 0.2}).add_to(m)

# Plot 2km buffer
folium.GeoJson(buffer_2km_wgs84.iloc[0], name='2km Buffer', style_function=lambda x: {'color': 'purple', 'fillOpacity': 0.1}).add_to(m)

# Plot houses not within 2km
for _, row in not_within_2km.iterrows():
    folium.GeoJson(row['geometry'], name='House', style_function=lambda x: {'color': 'red', 'fillOpacity': 0.4}).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'residential_houses_not_within_2km_of_library.html'
m.save(map_filename)

# Prepare summary
summary_rows = not_within_2km[['osid', 'description', 'geometry_area_m2']].head(5).to_dict('records')
summary = f"Found {len(not_within_2km)} residential houses in Exeter NOT within 2km of any library. First 5: {summary_rows}"

return [summary, map_filename, "Folium map of residential houses in Exeter NOT within 2km of any library, with libraries and 2km buffer shown.", map_filena
```

Tool code_executor returned result: - Found 12688 residential houses in Exeter NOT within 2km of any library. First 5: [{'osid': 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 99.338}, {'osid': '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'description': 'Mid-Terrace House', 'geometry_area_m2': 78.565}, {'osid': 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'description': 'Mid-Terrace House', 'geometry_area_m2': 66.91}, {'osid': 'a578be0a-7fff-4fbb-8ace-2cf755bbb53c', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 51.92}, {'osid': 'c86c6a23-5ced-4dae-be17-94f33e02c66d', 'description': 'Mid-Terrace House', 'geometry_area_m2': 51.91}]

Output from agent plotting_agent: Found 12,688 residential houses in Exeter NOT within 2km of any library. First 5: [{'osid': 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 99.338}, {'osid': '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'description': 'Mid-Terrace House', 'geometry_area_m2': 78.565}, {'osid': 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'description': 'Mid-Terrace House', 'geometry_area_m2': 66.91}, {'osid': 'a578be0a-7fff-4fbb-8ace-2cf755bbb53c', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 51.92}, {'osid': 'c86c6a23-5ced-4dae-be17-94f33e02c66d', 'description': 'Mid-Terrace House', 'geometry_area_m2': 51.91}]

Artifact name: residential_houses_not_within_2km_of_library.html

Artifact description: Folium map of residential houses in Exeter NOT within 2km of any library, with libraries and 2km buffer shown.

Artifact data: residential_houses_not_within_2km_of_library.html

---

host_agent

I have found all residential houses in Exeter that are NOT within 2km of any library. Here’s a summary of the results:

- Total residential houses in Exeter not within 2km of a library: 12,688
- Example house types include End-Of-Terrace and Mid-Terrace Houses.
- The map includes all such houses, the locations of libraries, and the 2km buffer zones around each library.

You can view the results in the artifact: **residential_houses_not_within_2km_of_library.html**, which visually displays the houses, libraries, and buffer zones.

Let me know if you need more details or want to explore specific areas or houses!